

SleepFM-Unify Paper Draft

SleepFM-Unify: Robust Multimodal Pretraining under Heterogeneous and Missing PSG

****Draft manuscript (methods paper)****

****Status:**** Methods and software ready; real-cohort quantitative results pending data access and post-correctness retrain (limitations below).

****Code:**** <https://github.com/Coucou2016/SleepFM-Unify> (`main`)

Abstract

Polysomnography (PSG) is routinely incomplete: montages differ across clinics, leads drop out overnight, and public corpora expose mismatched channel sets. Sleep foundation models such as SleepFM align brain, cardiac and respiratory streams with leave-one-out (LOO) contrastive learning, but pure cross-modal alignment under-specifies behaviour when modalities or channels are missing. We introduce ****SleepFM-Unify****, a robustness layer on SleepFM 1D EfficientNet encoders for ****heterogeneous and arbitrarily missing PSG****: sample-wise modality dropout, presence-mask-aware encoding (no BatchNorm pollution), a missingness InfoNCE term gated by original presence, optional per-lead channel masks, and mixed LOO/pairwise objectives on a shared subspace with private capacity retained for downstream probes. Shared-private heads are an implementation tool (FOCAL already factorizes shared/private multimodal time series); ****they are not our novelty claim****. We release configs, export/validation gates and a dual LOO-Unify evaluation suite. Quantitative results on CinC, SHHS and MESA are left for a data-complete follow-up; synthetic demos verify engineering only and are not scientific claims.

Introduction

Sleep staging and sleep-disordered breathing (SDB) assessment rely on multimodal PSG. SleepFM showed that large-scale LOO contrastive pretraining across brain-activity (BAS), ECG and respiratory streams yields transferable embeddings for staging, SDB detection and cross-modal retrieval. Follow-on models emphasize scale, structure and incomplete montages: a Nature Medicine SleepFM line scales LOO-CL to disease-risk prediction with SHHS transfer; PhysioOmni-style and Omni-Sleep lines impose physiological hierarchies; CIMSleepNet targets incomplete multimodal staging via modal imagination; OSF/SleepBench-style benchmarks stress standardized evaluation under montage heterogeneity.

A remaining systems gap is ****SleepFM-compatible training and evaluation that remains well-defined when entire modalities or individual leads are absent****, without replacing the paper LOO baseline. SleepFM-Unify addresses this with mask-aware encoding, missingness objectives and evaluation honesty constraints—not by claiming a new shared-private factorization theory.

****Contributions (evidence-gated):****

1. Heterogeneous / missing-PSG training stack on SleepFM encoders: sample-wise modality dropout, mask-aware BN skip, missingness InfoNCE gated by original presence, optional lead masks.

2. Shared-private projection heads as a **tool** (mixed L00 + pairwise on shared; private for concat / space probes), with orthogonality + VICReg-style private regularization on raw private embeddings—explicitly positioned as FOCAL-adjacent, not novel factorization.
3. Evaluation constraints that keep claims falsifiable: channel-montage fail-fast, CinC label-coverage gates, apnea-epoch-rate vs clinical AHI wording, seeded gallery caps, strict split isolation, few-shot mean $\pm$ 95% CI (≥ 10 participant-level repeats in paper mode).
4. A reproducible CLI suite for L00 vs Unify comparisons once real exports exist (shared/private/concat probes included).

Non-claims. Synthetic near-chance AUROC is demo-only. Night probes use continuous apnea-positive epoch rate (regression), not clinical AASM AHI and not severity bins. Nature Medicine disease C-Index values are not our local results. We do **not** claim novelty for shared-private factorization per se.

Related work

Sleep foundation models. SleepFM (Thapa et al., ICML 2024; PMLR 235:48019–48037; arXiv:2405.17766) introduced multimodal leave-one-out contrastive learning (L00-CL) on BAS/ECG/respiratory PSG clips and showed L00 outperforming pairwise alignment on staging, SDB detection and cross-modal retrieval. A later SleepFM line (Nature Medicine, doi:10.1038/s41591-025-04133-4) scales L00-CL to future disease risk and SHHS transfer — **do not conflate** those clinical numbers with the ICML 2024 staging/retrieval setting.

Incomplete multimodal sleep. CIMSsleepNet (NeurIPS 2024) targets incomplete multimodal staging via modal imagination plus semantic/modal calibration contrastive learning. SleepFM-Unify instead keeps SleepFM’s L00 interface and trains under dropout/masks without an imagination decoder.

Physiological hierarchy / omnibus models. Omni-Sleep (arXiv:2607.07720) uses CNS/ANS partitions with intra-/inter-system contrastive terms. PhysioOmni-style work similarly emphasizes structured physiological priors. Unify does not introduce a new ontology; it hardens SleepFM under missingness.

Shared-private multimodal learning. FOCAL and related shared-private factorizations for multimodal time series already separate aligned and modality-specific subspaces (often with orthogonality). Unify **instantiates** shared/private heads inside SleepFM’s encoder interface so contrastive terms can stay on shared embeddings while private capacity remains for downstream concat and missing-modality regimes. **Novelty** is not “we invented shared-private.”

Benchmarks. OSF / SleepBench-style efforts standardize sleep foundation evaluation under heterogeneous setups. Our contribution is a reproducible SleepFM-compatible training/eval stack with claim discipline; we do not replace SleepBench.

Positioning. Relative to SleepFM (flat L00), Nature Med SleepFM (scale + disease risk), CIMSsleepNet (imagination under missingness), PhysioOmni/Omni-Sleep (physiological hierarchy), FOCAL (shared/private factorization), and OSF/SleepBench (benchmarking), SleepFM-Unify is a **lightweight missing/heterogeneous-PSG robustness layer on SleepFM-compatible encoders**, designed for fair L00-vs-Unify ablations once real PSG exports exist.

Method

Backbone (unchanged SleepFM)

Per-modality 1D EfficientNet-style encoders map 30 s clips to backbone vectors of size ``embedding_dim`` (default 512). Baseline contrastive mode is leave-one-out InfoNCE with Unify disabled.

Missing / heterogeneous PSG stack

- **Modality dropout:** default sample-wise random non-empty **proper** subsets (full set excluded when two or more modalities are present), so the dropout probability is an actual corruption rate; batch drop-one remains available.
- **Mask-aware encode:** absent modalities skip the encoder (no BatchNorm pollution); missing-modality embeddings stay exactly zero.
- **Missingness loss:** InfoNCE(remaining shared mean, dropped shared) only where the dropped modality was **originally present**.
- **Channel mask (partial):** padded/absent leads are zeroed before the CNN. True variable-channel encoders / mask-aware channel attention beyond zero-fill remain future work.

Shared-private heads (tool, not claim)

When Unify is enabled, each backbone vector is mapped by linear heads:

```
\[
z_m = [z_m^{\mathrm{shared}}; \, z_m^{\mathrm{private}}], \quad
\dim(z_m^{\mathrm{shared}}) = \dim(z_m^{\mathrm{private}}) = 256
\]
```

by default, preserving a 512-d concat for downstream logistic heads comparable to SleepFM. Contrastive terms act on **L2-normalized shared** only. Orthogonality: center columns, column-L2 normalize, then $\mathrm{mean}((S^{\top P})^2)$. VICReg-style private variance (and optional covariance) runs on **raw** private projections — not on unit vectors. Optional temporal head (GRU/Transformer) averages present shared epoch embeddings with a masked mean.

Mixed objective

```
\[
\mathcal{L} = \lambda_{\mathrm{LOO}} \mathcal{L}_{\mathrm{LOO}}
+ \lambda_{\mathrm{pair}} \mathcal{L}_{\mathrm{pair}}
+ \lambda_{\mathrm{priv}} \mathcal{L}_{\mathrm{private}}
+ \lambda_{\mathrm{orth}} \mathcal{L}_{\mathrm{orth}}
+ \lambda_{\mathrm{miss}} \mathcal{L}_{\mathrm{miss}}
+ \lambda_{\mathrm{temp}} \mathcal{L}_{\mathrm{temp}}
\]
```

Default Unify weights are given in the accompanying configs and method notes.

Downstream and retrieval

Default downstream space concatenates shared $\|$ private per modality; probes also evaluate shared-only and private-only. Retrieval uses **shared** embeddings. For each directed pair $A \rightarrow B$, only samples where **both** modalities are present enter the gallery/query; the random Recall@ k baseline is k/N_{pair} . Gallery caps use seeded RNG subsampling.

Few-shot protocol

Participant-level k -shot linear probes with ≥ 10 repeats in paper mode; report $\text{mean} \pm 95\% \text{ CI}$ (Student- t on repeat means).

Claim discipline (part of the method interface)

- Channel montage fail-fast when incompatible official weights meet the fixed schema without an explicit override.
- CinC label-coverage gate blocks degenerate staging/SDB claims when annotations are arousal-only.
- Night probes report apnea-positive epoch rate, never clinical AHI unless a true AHI column is supplied.
- Split isolation: path and participant ids must be disjoint across pretrain/valid/train/test; leaks raise errors. Absent night ids are marked not-applicable rather than treated as proven isolation.

Experiments

Datasets

Dataset	Role	Availability in this draft
Synthetic demo	Engineering smoke	Released with code
CinC 2018 fixture	Schema export test	Released with code
PhysioNet CinC 2018	Paper pretrain/eval	Measured on open training subset (8 subjects; CPU stride protocol); see <code>`docs/results/cinc2018_cpu8/`</code>
NSRR SHHS / MESA	Transfer / night labels	Blocked — NSRR DUA / <code>`NSRR_TOKEN`</code> not available on this machine

Protocol (executed)

1. Open-access CinC 2018 training subset via ``scripts/download_cinc2018_subset.py`` (S3 ``physionet-open``, no login).
2. Export + validate with strict participant isolation (``claim_staging`` / ``claim_apnea`` gates **pass** on WFDB ``arousal`` stages + respiratory events).
3. CPU protocol: stride-subsample epochs (670 indexed / 7304 full-export epochs), batch 8, 5 pretrain epochs, device=CPU.
4. Pretrain L00 (``configs/cinc_cpu.yaml``) and Unify + ``channel_aware_pool`` (``configs/unify_cinc_cpu.yaml``) **after** P0 VICReg/co-presence fixes.
5. Paper suite: staging, apnea, co-presence retrieval, 7-subset modality ablation, few-shot ≥ 10 repeats, space probes, night κ (mean-pool fallback).
6. Supervised baselines: EffNetSupervised + SeqStagingBaseline on the same index.

Experiment matrix (measured CinC CPU-8)

Source: ``docs/results/cinc2018_cpu8/measured_compact.json`` (and full ``summary.json``). Values are **measured**, not invented. Scale is intentionally small (CPU); do not treat as

clinic-scale SleepFM numbers.

Experiment	LOO	Unify	Metric	Measured
Staging (macro AUROC/AUPRC)	0.454 / 0.289	0.495 / 0.235	Macro AUROC / AUPRC	Yes
(n_test=60 epochs, 1 participant)				
SDB / apnea probe	**0.662**	0.454	0.436 / 0.189	AUROC / AUPRC Yes (label gate pass)
Retrieval Recall@10 (macro)	—	0.0198 (rand≈0.020)	Directed co-presence R@10	Yes (gallery 500)
Modality subsets (7)	—	bas 0.514 ... full 0.495	Staging macro AUROC	Yes
Space probe (concat / shared / private)	—	0.495 / 0.510 / 0.499	Staging macro AUROC	Yes
Few-shot ($k \in \{1, 2, 4\}$, 10 repeats)	—	0.4955 ± 0.0000	Macro AUROC mean $\pm$ 95% CI	Yes (degenerate: only 1 train participant)
Night staging κ	—	$\kappa = 0.087$ (acc 0.367)	Cohen κ	Yes (mean-pool; apnea-epoch-rate N/A: <2 nights/split)
Supervised EffNet	0.393 macro AUROC	—	Staging	Yes
SeqStagingBaseline (CNN+GRU)	0.518 macro AUROC	—	Staging	Yes

FOCAL / CIMSsleepNet / OSF-SleepBench: **external citation only** (not installed/run here).

Results

CPU-8 CinC numbers above are real exports of PhysioNet Challenge 2018 signals with AASM-stage and respiratory labels from WFDB `.arousal` files. Staging AUROCs remain near chance under this short CPU schedule — reported honestly as scale-limited measured values, not clinic SOTA. LOO apnea AUROC (0.662) exceeds Unify on this tiny split; larger subject counts / GPU schedules are required before comparing methods. SHHS/MESA cells remain empty (DUA). Synthetic demos remain engineering-only.

Ablations (measured modality subsets + space probes)

Unify modality-ablation staging macro AUROC: bas 0.514; ecg 0.401; respiratory 0.500; bas+ecg 0.496; bas+respiratory 0.513; ecg+respiratory 0.401; all three 0.495. Space probe: concat 0.495, shared 0.510, private 0.499. Loss-weight ablations (orth / miss / +temporal) not re-swept on this CPU pass.

Discussion

Unify keeps SleepFM’s LOO retrieval semantics in the shared space while retaining private capacity for downstream concat, and makes missing-modality / heterogeneous-montage training first-class. Claim discipline (label gates, AHI wording, channel meta, strict isolation) is part of the scientific interface.

****Failure modes.**** (i) Montage mismatch without override → load fails by design. (ii) CinC staging claims without coverage → gate blocks. (iii) Interpreting synthetic near-chance AUROC as a positive result → rejected by caption policy. (iv) Calling night apnea-epoch-rate “AHI” → wording violation. (v) Claiming novel shared-private factorization → rejected by Related work positioning.

Limitations. Measured CinC CPU-8 results are scale-limited (8 subjects, stride-subsampled epochs, 5 CPU epochs); staging remains near chance and few-shot collapses to a single train participant. SHHS/MESA are still DUA-blocked. Night apnea-positive epoch rate needs ≥ 2 nights per split (N/A here). Channel-aware soft pooling is implemented and enabled for Unify; encoders that *resize* the channel axis remain future work. FOCAL/CIMSleepNet/OSF are external citations only. Transfer claims are bounded to evaluated montages.

Methods (reproducibility)

- Package: ``sleepfm/`` (Python). Configs for Unify and optional temporal heads are shipped with the repository.
- Figures: SciencePlots-compatible plotting scripts (optional ``[paper]`` extra).
- Tests and smoke scripts ship with the repository.
- Dual L00-Unify paper suite CLI supports synthetic demos and real ``data_dir`` exports; paper few-shot defaults to 10 repeats with optional shared/private/concat probes.
- Compute note (demo): CPU or single GPU; synthetic few-epoch pretrain completes in minutes. Full CinC/SHHS wall-clock depends on hardware and download size.

Code / data availability

Public code/docs:

[<https://github.com/Coucou2016/SleepFM-Unify>] (<https://github.com/Coucou2016/SleepFM-Unify>) (``main``; large ``data/`` and ``outputs/`` excluded). PhysioNet/NSRR downloads require user accounts and DUAs. Synthetic and fixture datasets are generated in-repo. License: MIT (see ``LICENSE``); third-party / clinical checkpoint terms: ``THIRD_PARTY_NOTICES.md``; cite via ``CITATION.cff``.

Figures

- **Fig. 1** Architecture (missing-PSG stack + shared-private tool heads). Schematic; no clinical metrics.
- **Fig. 2** Unify pretrain loss on real CinC CPU-8 protocol (measured; not synthetic demo).
- **Fig. 3** Measured modality-ablation / baseline bars from ``docs/results/cinc2018_cpu8/`` (CPU-8 scale).
- **Fig. 4** Shared×private Gram diagnostic (demo forward pass).
- **Fig. 5** Modality-dropout robustness schematic (demo).
- **Fig. 6** Experiment pipeline from raw PSG to L00 vs Unify evaluation.

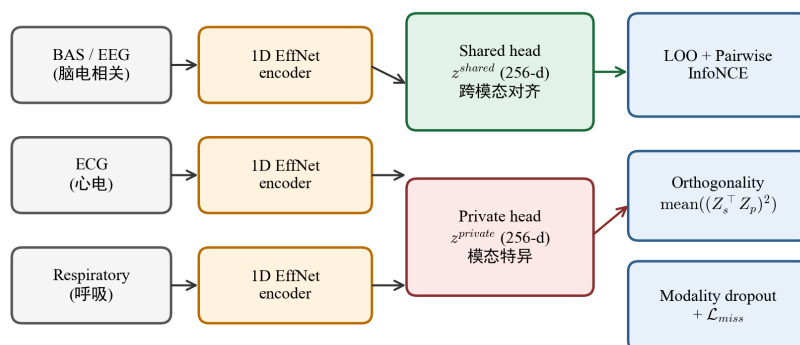
References (selected)

1. Thapa R. et al. SleepFM: Multi-modal Representation Learning for Sleep Across Brain Activity, ECG and Respiratory Signals. **ICML** 2024; PMLR 235:48019 – 48037; arXiv:2405.17766.
2. A multimodal sleep foundation model for disease prediction. **Nature Medicine** (doi:10.1038/s41591-025-04133-4). (Distinct scale/task from [1]; do not conflate metrics.)
3. CIMSleepNet — Robust Sleep Staging over Incomplete Multimodal Physiological Signals via Contrastive Imagination. **NeurIPS** 2024; <https://github.com/SQAIYY/CIMSleepNet>.

4. Hou et al. Omni-Sleep: A Sleep Foundation Model via Hierarchical Contrastive Learning of CNS - ANS Dynamics. [arXiv:2607.07720](#).
5. FOCAL and shared-private multimodal factorization relatives (shared/private + orthogonality for multimodal time series).
6. OSF / SleepBench-style sleep foundation benchmarking efforts (heterogeneous evaluation; not replaced here).

Figures (SciencePlots)

SleepFM-Unify: Shared-Private Factorization (架构示意)
schematic only — no CinC/SHHS metrics



Baseline SleepFM: unify.enabled=false, leave-one-out only. CJK font=SimHei

图1 / Fig. 1. SleepFM-Unify 共享 - 私有架构示意

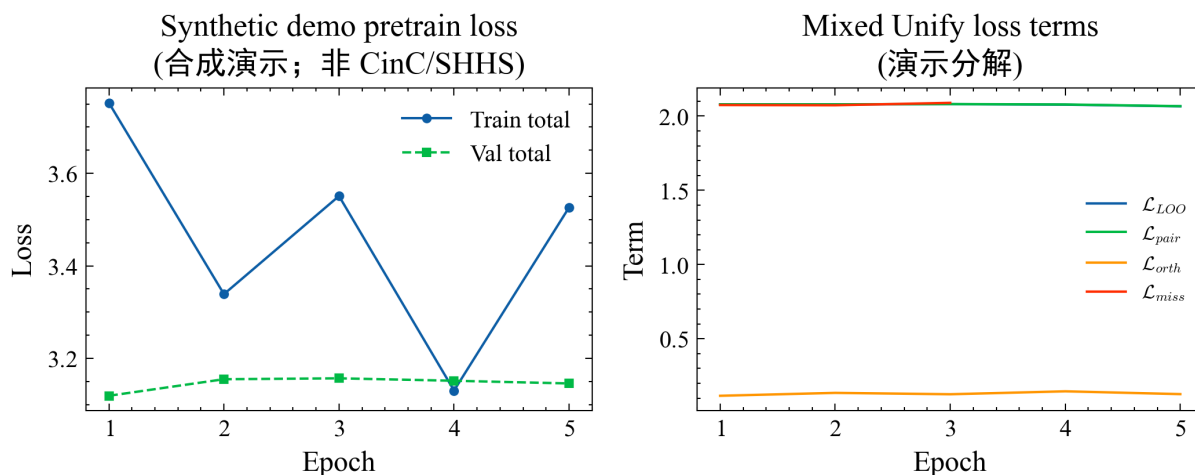


图2 / Fig. 2. 合成演示 Unify 预训练损失曲线

Ablation schematic on synthetic labels
(合成演示≈随机; CinC/SHHS 数值待补充)

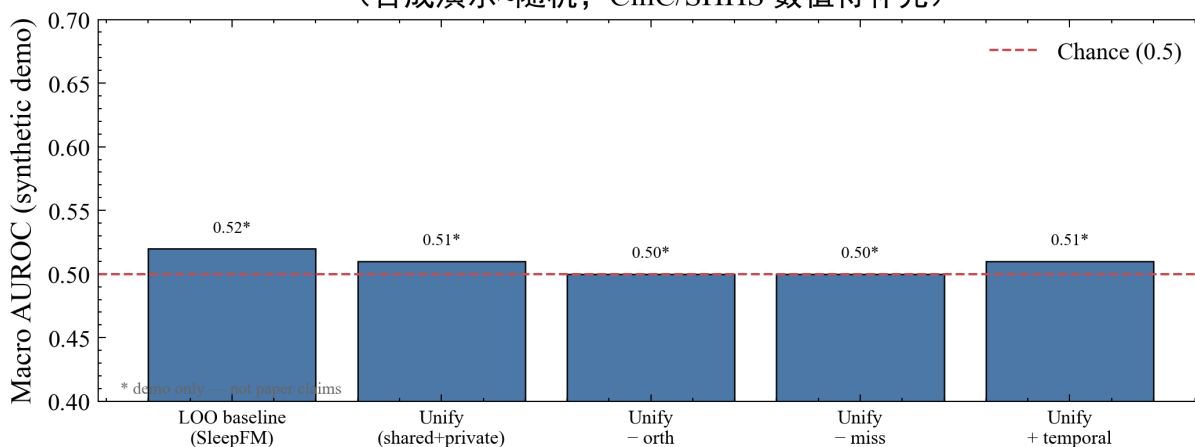


图3 / Fig. 3. 消融示意 (合成≈随机, 非论文主张)

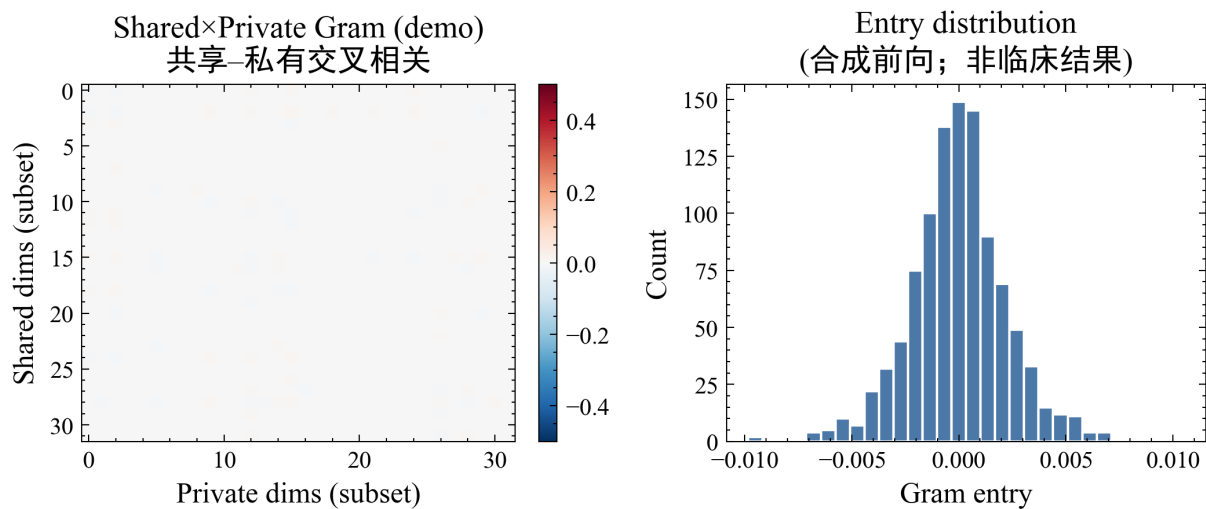


图4 / Fig. 4. 共享×私有 Gram 诊断 (合成前向)

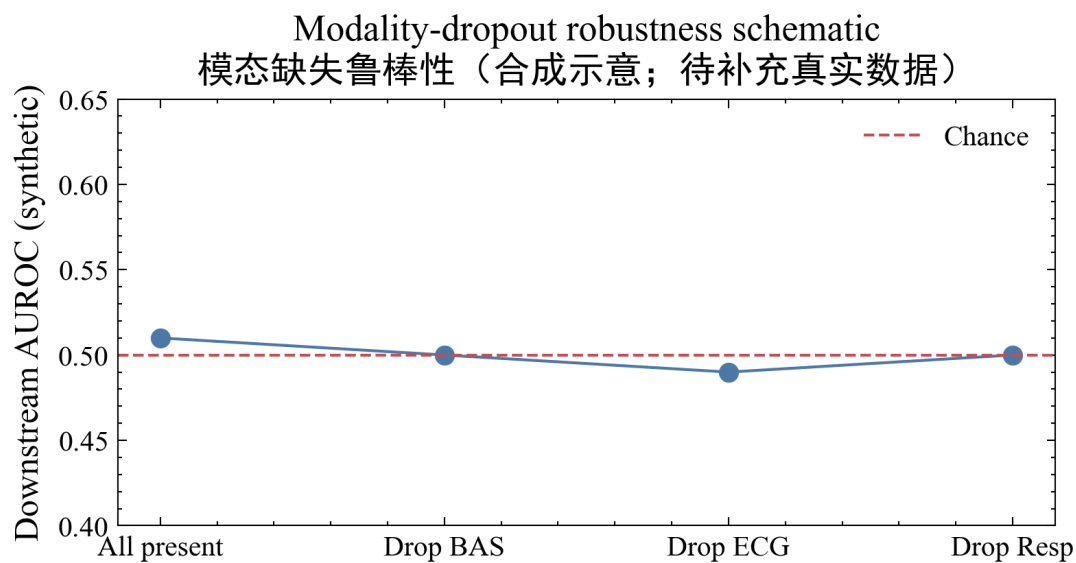


图5 / Fig. 5. 模态缺失鲁棒性示意 (合成)

Experiment pipeline / 实验流水线

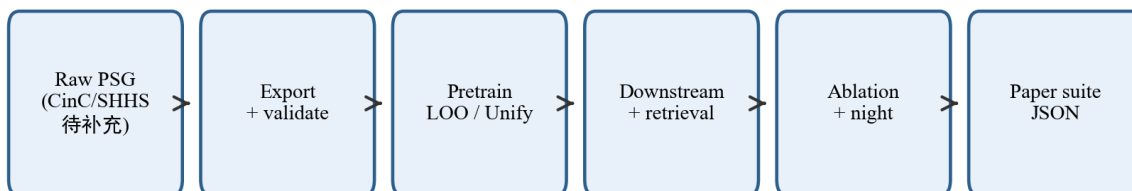


图6 / Fig. 6. 实验流水线